Department of Computer Science



Programming Fundamentals Lab

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Task - 1

Assure that the following statement are defined:

```
int age;
double pay;
char section;
```

Write a single cin statement that will read input into each of these variables.

```
#include <iostream>
using namespace std;
int main()
{
    int age;
    double pay;
    char section;
cout<<"enter age, pay, section"<<endl;
cin>>age>>pay>>section;
return 0;
}
```

Output:

```
Enter age, pay, section
19
1000
A
```

```cpp
#include <iostream>
using namespace std;
int main ( )
{
    int age;
    double pay;
    char section;
    cout<<"enter age,pay,section"<<endl;
    cin>>age>>pay>>section;
    return 0;
}
```



```
enter age,pay,section
19
800
A

[Program finished]
```

Task-2

Complete the following table by writing the value of each expression in the value column according C++ language rules.

Expression	Value
28/4-2	
6+12*2-8	
4+8*2	
6+17%3-2	
2+22*(9-7)	
(8+7)*2	
(16+7)%2-1	
12/(10-6)	
(19-3)*(2+2)/4	

```
#include <iostream>
using namespace std;
int main()
{
    cout<<"28/4-2"<<endl;
    cout<<"6+12*2-8"<<endl;
    cout<<"4+8*2"<<endl
    cout<<"6+17%3-2"<<endl;
    cout<<"2+22*(9-7)"<<endl;
    cout<<"(8+7)*2"<<endl;
    cout<<"(16+7)%2-1"<<endl;
    cout<<"12/(10-6)"<<endl;
```

```
cout << "(19-3) * (2+2) is " << endl;
return 0;
}
```

Output:

```
5
22
20
6
46
30
0
3
16
```

```cpp
#include <iostream>
using namespace std;
int main ( )
{
    cout << (28 / 4 - 2) << endl;
    cout << (6 + 12 * 2 - 8) << endl;
    cout << (4 + 8 * 2) << endl;
    cout << (6 + 17 % 3 - 2) << endl;
    cout << (2 + 22 * (9 - 7)) << endl;
    cout << ((8 + 7) * 2) << endl;
    cout << ((16 + 7) % 2 - 1) << endl;
    cout << (12 / (10 - 6)) << endl;
    cout << ((19 - 3) * (2 + 2) / 4) << endl;
return 0;
}
```



```
5
22
20
6
46
30
0
3
16

[Program finished]
```

Task - 3

Assume a program has the following variable definations:

```
int units;
float mass;
double weight;
weight = mass * units;
```

Which automatic data type convession will take place?

A. mass is demoted to an int, units remains an int, and results of mass * units is an int?

B. units is promoted to a float, remains a float and result of mass * units is a float.

C. units is promoted to a float, mass remains a float, & result of mass * units is

a double.

```
#include <iostream>
using namespace std
int main ()
{

    int units = 4;
    float mass = 10.5f;
    double weight;
    // Multiplying float and int perform
                        promotion.
    weight = mass * units;
    cout << "The calculated weight is:" << weight << endl;
    return 0;
}
```

Output:

The calculated weight is: 42

```cpp
#include <iostream>

using namespace std;

int main() {
    int units = 4;
    float mass = 10.5f;
    double weight;


    weight = mass * units;


    cout << "The calculated weight is: "
<<weight<< endl;

    return 0;
}

```


The calculated weight is: 42
[Program finished]

Task-4

Assume a program has the following variable definations:

int a, b=2; , float c= 4.2;

and the following statement: a = b * c;

A. 8.4 B. 8 C. 0 D. None of the above

```
#include <iostream>
using namespace std;
int main ( )
{
```

```
    int a, b= 2;
    float c = 4.2;
cout << "The value stored in a is:" << a<<
endl;
    return 0;
}
```

Output:

The value stored in a is: 8

```cpp
#include <iostream>

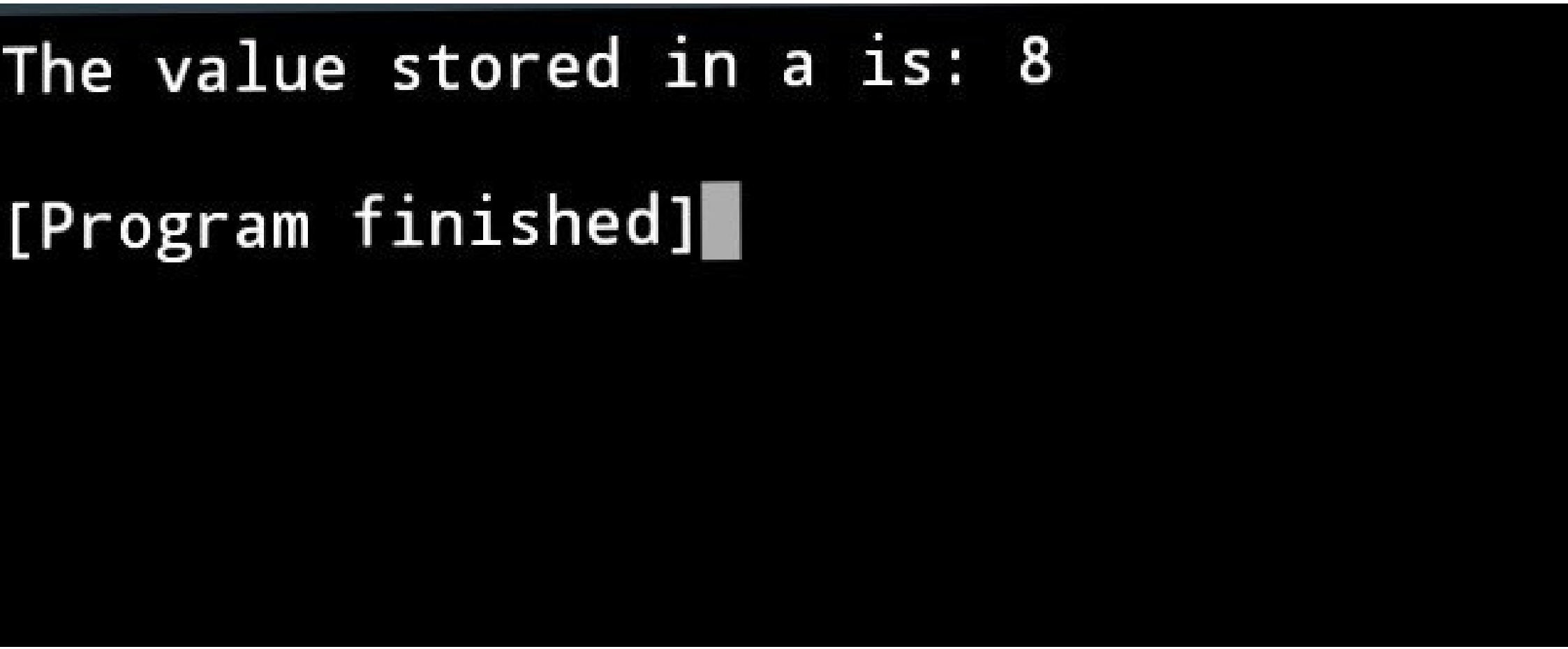
using namespace std;

int main() {

    int a, b = 2;
    float c = 4.2;
     a = b * c;
    cout << "The value stored in a is: " << a <<
endl;

    return 0;
}

```


The value stored in a is: 8
[Program finished]

TASK - 5

Assume that qtyt qty and salesReps are both integer. Use a type cast expression to rewrite the following statement so it will no longer perform division integer.

```
#include <iostream>
using namespace std;
int main() {

  int qty = 26;
  int salesReps = 4;
  double each units;
  unitEach = static_cast<double>(qty) / salesReps;
cout<<" Quantity:"<< qty <<endl;
cout<<" Sales Reps:"<< salesReps<< endl;
cout<<" Units Each:"<< unitReps <<endl;
return 0;
}
```

Output:

```
Quantity: 26
Sales Reps: 4
Units Each: 6.5
```

```cpp
#include <iostream>

using namespace std;

int main() {
    int qty = 25;
    int salesReps = 4;
    double unitEach;
    unitEach = static_cast<double>(qty) /
salesReps;

    cout << "Quantity: " << qty << endl;
    cout << "Sales Reps: " << salesReps << endl;
    cout << "Units Each : " << unitEach << endl;

    return 0;
}
```

```
Quantity: 25
Sales Reps: 4
Units Each : 6.25

[Program finished]
```

Task-6

Write a program that can be used as math tutor for a young student. The Program should display two random numbers to be added, such as

```
  247
 +129
-----
```

The program should then pause while the student works on the problem. When the student is ready to to check the answer, he or she can press a key and the program will display the correct solution

```
  247
 +129
-----

  376
```

```
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;
int main()
{
  srand(time(0));
 int num1=rand()%900+100;
 int num2=rand()%900+100;
 cout<<num1<<endl;
 cout<<"+"<<num2<<endl;
 cout<<"-----"<<num2<<endl;
 cin.get();
```

```
    cout << num1 << endl;
    cout << " + " << num2 << endl;
    cout << " ------ " << endl;
    cout << (num1 + num2) << endl;
    return 0;
}
```

Output:

```
  456
 + 844
-------
 1300
```

```cpp
#include <iostream>
#include <cstdlib>
#include <ctime>

using namespace std;

int main() {
    srand(time(0));

    int num1 = rand() % 900 + 100;
    int num2 = rand() % 900 + 100;

    cout << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "-------" << endl;

    cin.get();

    cout << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "-------" << endl;
    cout << (num1 + num2) << endl;

    return 0;
}

```

```
454
+ 848
--------
300
454
+ 848
--------
1302

[Program finished]
```

Home-Task

Task-1

Each of the following program has some errors. Locate as many as you can.

Program 1

```
#include using namespace std;
void main ()
{

double number 1 , number 2, sum:
cout << "Enter a numbers:";
cin << number 1;
cout << "Enter another number:";
cin << number 2;
number 1 + number 2 = sum;
cout "The sum of the two numbers is" << sum
}
```

```
#include <iostream>
using namespace std;
int main ()
{
  double number1, number2, sum;
  cout << "Enter a number";
  cin >> number1;
  cout << "Enter another numbers";
  cin >> number2;
  sum = number1 + number2;
cout << "The sum of the two numbess:" <<
  sum;
  return 0;
}
```

Output:

Enter a number: 650

Enter another number: 430

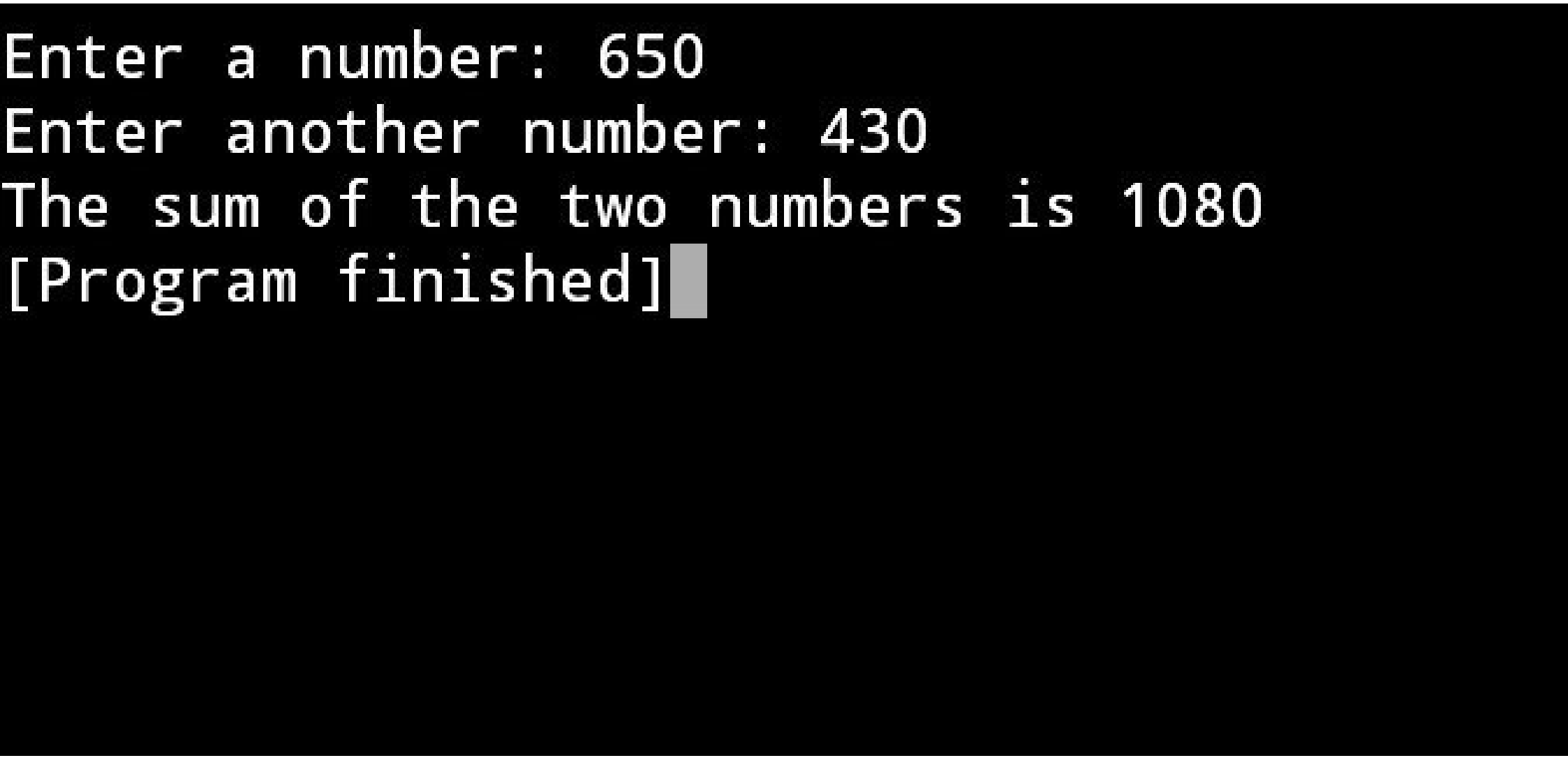
The sum of two numbess is 1080

```cpp
#include <iostream>

using namespace std;

int main()
{
    double number1, number2, sum;
    cout << "Enter a number: ";
    cin >> number1;
    cout << "Enter another number: ";
    cin >> number2;
    sum = number1 + number2;
    cout << "The sum of the two numbers is " <<
sum;

    return 0;
}
```


Enter a number: 650
Enter another number: 430
The sum of the two numbers is 1080
[Program finished]

Program-2

```
#include <iostream>
using namespace std;
void main()
{
  int number1, number2;
  float qoutient;
 cout<<"Enter two numbers and I will
divide \n";
cout<<" the first by the second for you.\n";
 cin>> number1, number2;
qvotient =float<static_cast>(number1)/
          number2;
cout << qvotient;
 }
```

```
#include <iostream>
using namespace std;
int main()
{
  int number1, number2;
  float qvotient;
cout<<"Enter two numbers and I will divide\n";
cout<<"the first by the second for you.\n";
   cin >> number1 >> number2;
qvotient = static_cast<float>(number1)/number2;
  cout << qvotient;
```

```
return 0;
}
```

Output:

```
Enter two numbers and I
will divide the first by
second for you.
45
9
5
```

```cpp
#include <iostream>

using namespace std;

int main()
{
    int number1, number2;
    float quotient;
    cout << "Enter two numbers and I will divide\n";
    cout << "the first by the second for you.\n";
    cin >> number1 >> number2;
    quotient = static_cast<float>(number1) / number2;
    cout << quotient;

    return 0;
}
```

```
Enter two numbers and I will divide
the first by the second for you.
45
9
5
[Program finished]
```

Task - 2

Average of values to get average of a series of sales, you add the values up and then divide the sum by the numbers of values. Write a program that stores the following values in five different variables: 28, 32, 37, 24, and 33. The program should first calculate the sum of these five variables and store the result in a separate variable named sum. Then, the program should divide the sum variable by 5 to get the average. Display the average on the screen.

```
#include <iostream>
using namespace std;
int main ()
{
    double num1 = 28, num2 = 32, num3 = 37,
        num4 = 24, num5 = 33;
    double sum, average;
sum = num1 + num2 + num3 + num4 + num5;
```

```
    averge = sum / 5;
    cout << average;
    return 0;
}
```

Output:

30.8

```cpp
#include <iostream>

using namespace std;

int main()
{
    double num1 = 28, num2 = 32, num3 = 37, num4 = 24, num5 = 33;
    double sum, average;

    sum = num1 + num2 + num3 + num4 + num5;
    average = sum / 5;

    cout << average;

    return 0;
}
```

```
30.8
[Program finished]
```